



Name: Gargi Sukhatankar
Roll No: B027

SAP ID: 60019220125
Branch: CSE(ICB)

EXPERIMENT 7

Aim: To compare different types of Taggers.

1. Rule-Based Tagger

- **Theory:** Uses predefined linguistic rules to assign POS tags.
- **Pros:** Works well with small datasets, interpretable.
- **Cons:** Struggles with ambiguity, requires manual rule creation.
- **Example:** Brill Tagger

```
import nltk
from nltk.tokenize import word_tokenize
from nltk.tag import RegexpTagger

# Define regex-based tagging rules
patterns = [
    (r'.*ing$', 'VBG'), # gerunds
    (r'.*ed$', 'VBD'), # past tense verbs
    (r'.*es$', 'VBZ'), # 3rd person singular present
    (r'\b\w+\b', 'NN') # nouns (default)
]

tagger = RegexpTagger(patterns)
sentence = word_tokenize("This is Vipul Mhatre from TE CSE(ICB) Department")
print(tagger.tag(sentence))
```

```
[('This', 'NN'), ('is', 'NN'), ('Vipul', 'NN'), ('Mhatre', 'NN'), ('from', 'NN'), ('TE', 'NN'), ('CSE', 'NN'),
```

2. Statistical Tagger

- **Theory:** Uses probabilistic models (like Hidden Markov Models) to predict POS tags based on training data.
- **Pros:** More accurate than rule-based methods, learns from data.
- **Cons:** Requires a large labeled dataset, can be biased by training data.



Name: Gargi Sukhatankar
Roll No: B027

SAP ID: 60019220125
Branch: CSE(ICB)

- **Example: HMM Tagger**

```
import nltk
nltk.download('averaged_perceptron_tagger')
nltk.download('treebank')
nltk.download('punkt')
```

```
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data] /root/nltk_data...
[nltk_data] Package averaged_perceptron_tagger is already up-to-
[nltk_data] date!
[nltk_data] Downloading package treebank to /root/nltk_data...
[nltk_data] Unzipping corpora/treebank.zip.
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data] Package punkt is already up-to-date!
True
```

```
from nltk.tag import hmm
from nltk.corpus import treebank
```

```
# Load treebank data
```

```
train_data = treebank.tagged_sents()[0:3000]
test_data = treebank.tagged_sents()[3000:]
```

```
# Train HMM Tagger
```

```
trainer = hmm.HiddenMarkovModelTrainer()
tagger = trainer.train(train_data)
```

```
# Test the model
```

```
print("Statistical Tagging Accuracy:", tagger.evaluate(test_data))
```

```
sentence = "This is Vipul Mhatre from TE CSE(ICB) Department".split()
print("HMM Tagging:", tagger.tag(sentence))
```

```
<ipython-input-8-9cfac70aad41>:13: DeprecationWarning:
  Function evaluate() has been deprecated. Use accuracy(gold)
  instead.
  print("Statistical Tagging Accuracy:", tagger.evaluate(test_data))
/usr/local/lib/python3.11/dist-packages/nltk/tag/hmm.py:333: RuntimeWarning: overflow encountered in cast
  X[i, j] = self._transitions[si].logprob(self._states[j])
/usr/local/lib/python3.11/dist-packages/nltk/tag/hmm.py:335: RuntimeWarning: overflow encountered in cast
  O[i, k] = self._output_logprob(si, self._symbols[k])
/usr/local/lib/python3.11/dist-packages/nltk/tag/hmm.py:331: RuntimeWarning: overflow encountered in cast
  P[i] = self._priors.logprob(si)
/usr/local/lib/python3.11/dist-packages/nltk/tag/hmm.py:363: RuntimeWarning: overflow encountered in cast
  O[i, k] = self._output_logprob(si, self._symbols[k])
Statistical Tagging Accuracy: 0.36844377293330455
HMM Tagging: [('This', 'DT'), ('is', 'VBZ'), ('Vipul', 'NNP'), ('Mhatre', 'NNP'), ('from', 'NNP'), ('TE', 'NNP')]
```

3. Neural Network-Based Tagger



Name: Gargi Sukhatankar
Roll No: B027

SAP ID: 60019220125
Branch: CSE(ICB)

- **Theory:** Uses deep learning models (like LSTMs, Transformers) to tag words based on context.
- **Pros:** Highly accurate, handles complex dependencies.
- **Cons:** Computationally expensive, requires large data.
- **Example:** BERT-based POS Tagger

```
import spacy

# Load SpaCy model
nlp = spacy.load("en_core_web_sm")
sentence = "This is Vipul Mhatre from TE CSE(ICB) Department"

# Perform POS tagging
doc = nlp(sentence)
for token in doc:
    print(token.text, token.pos_)
```

```
This PRON
is AUX
Vipul PROPN
Mhatre PROPN
from ADP
TE PROPN
CSE(ICB PROPN
) PUNCT
Department NOUN
```

Comparison Table:



SHRI VILEPARLE KELAVANI MANDAL'S
DWARKADAS J. SANGHVI COLLEGE OF ENGINEERING
(Autonomous College Affiliated to the University of Mumbai)
NAAC ACCREDITED with "A" GRADE (CGPA : 3.18)



Name: Gargi Sukhatankar
Roll No: B027

SAP ID: 60019220125
Branch: CSE(ICB)

Feature	Rule-Based Tagger	Statistical Tagger	Neural Network Tagger
Accuracy	Low to Moderate	Moderate to High	Very High
Training Needed	None	Requires Labeled Data	Requires Large Labeled Data
Complexity	Simple	Moderate	High
Interpretability	High	Moderate	Low
Handling Ambiguity	Poor	Moderate	Excellent

Conclusion: Thus we successfully compared different types of Taggers.